Lecture 2
Simulink Examples

ME 132 - Dynamic Systems & Feedback

GSI Introductions

Ryan Park

2nd year PhD under Negar Mehr

Research topics: Game theory, optimization, uncertainty, robotics

Discussions/Lab: Th 9-10, 10-11, 1109 Etcheverry Hall

OH: Th 1-2, 1171 Etcheverry Hall

GSI Introductions

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3rd year PhD under Prof. Negar Mehr

Research topics: Robot Learning, LLMs, Reinforcement Learning

Discussion/Lab: Tue 11-12, 12-1, 1109 Etcheverry Hall

OH: Wed 11-12, 1171 Etchevery Hall

Logistics

- Download MATLAB + Simulink (R2026a) until your next discussion session.
- First four discussion sessions are going to be about Simulink. Then, we will move on to solving homework problems.
- Please attend the discussion sessions you are registered in.
- Please contact us through EdStem!

Roadmap

- How should we think about control systems?
- How can we simulate these systems for experimentation?
- What tools exist in industry and academia?

Control Systems

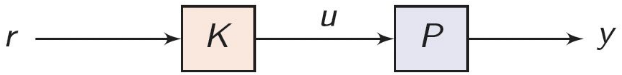
P: Plant, System

y: The system’s output or measurable signals, i.e. temperature of a room, the GPS coordinates of a car

K: Controller

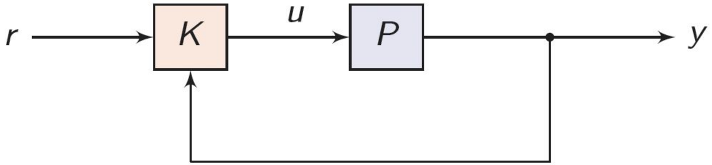
r: The controller’s reference, i.e. temperature for an air conditioner, destination for an autonomous vehicle

u: The controller’s output, i.e. voltage signal from the AC buttons, accelerator/brake/steering commands



“Open Loop”, “Feedforward Control”

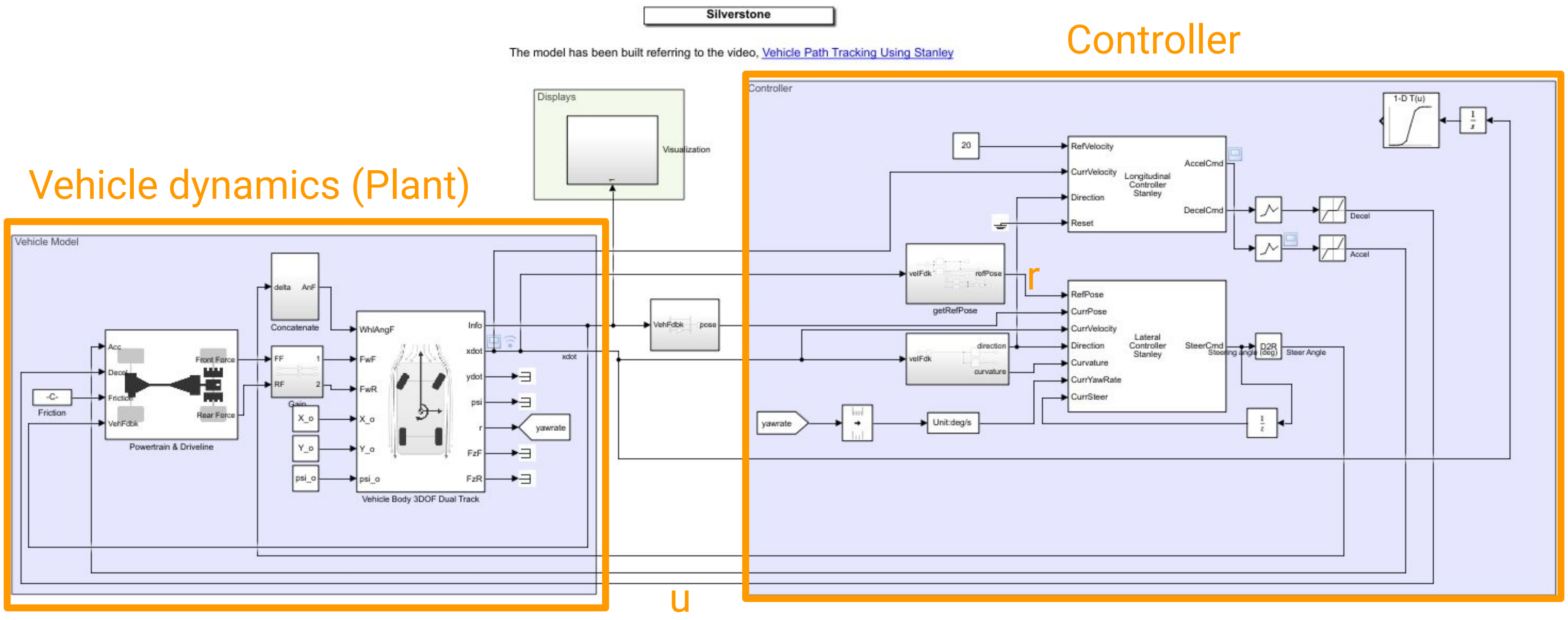
- Simplest model possible - controller just transforms r into u.
- If the controller does not perfectly predict what P will do, then the system may break.



“Closed Loop”, “Feedback Control”

- A complex model - controller transforms r into u, knowing the effect of previous u.
- The controller can *adapt* to unexpected behavior by seeing y.

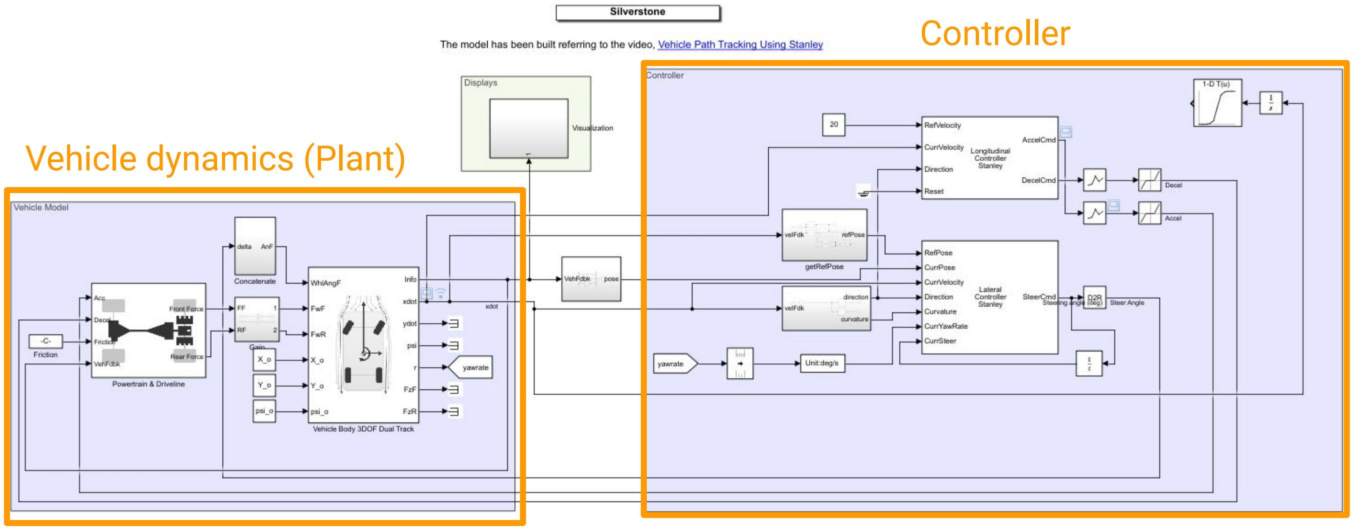
Realistic Systems – Car

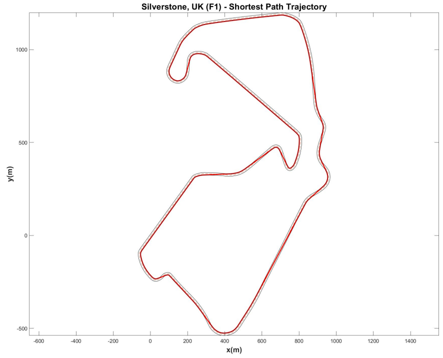


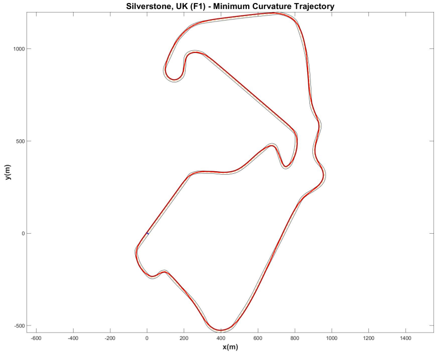
MW208 Raceline Optimization: https://github.com/putta54/MW208_Raceline_Optimization/tree/94e9a6e9fb23d14f9718b5472682fb379cc6c130?tab=readme-ov-file

MATLAB Simulink

Simulink is a graphical tool for modeling dynamic systems (plant), controller, and sensors





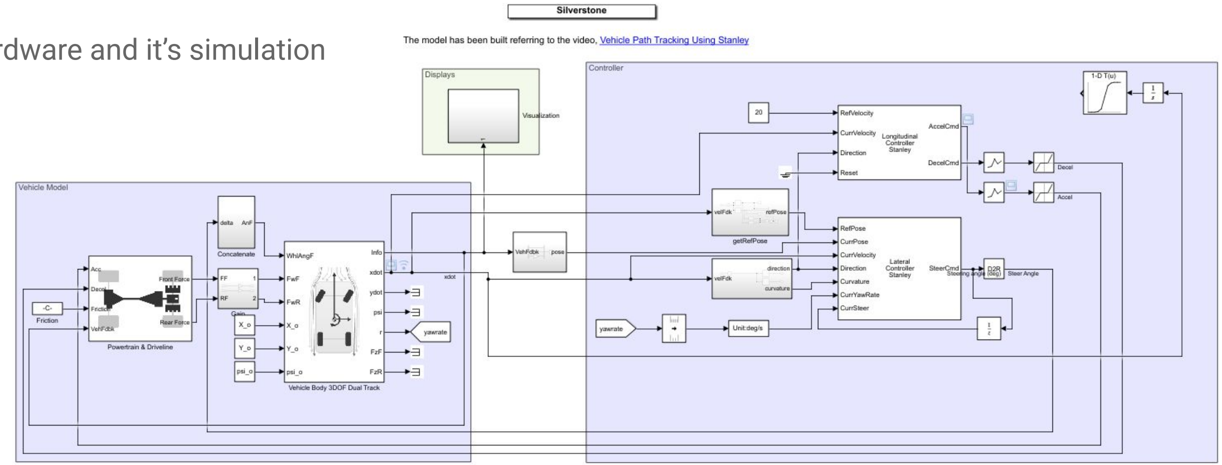


MW208 Raceline Optimization: https://github.com/putta54/MW208_Raceline_Optimization/tree/94e9a6e9fb23d14f9718b5472682fb379cc6c130?tab=readme-ov-file

MATLAB Simulink

Key features

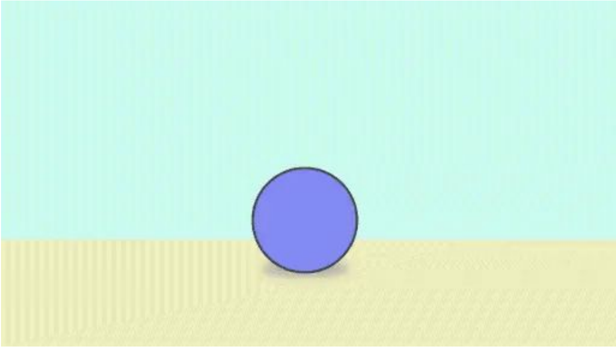
- Block Diagram Modeling
  - Intuitive drag and drop of blocks instead of writing long equations or code
- Numerical Simulation of Dynamic Systems
  - Time-based simulation solver for continuous, discrete, or hybrid systems
- Visualization
  - Easy visualization of realistic hardware and it’s simulation

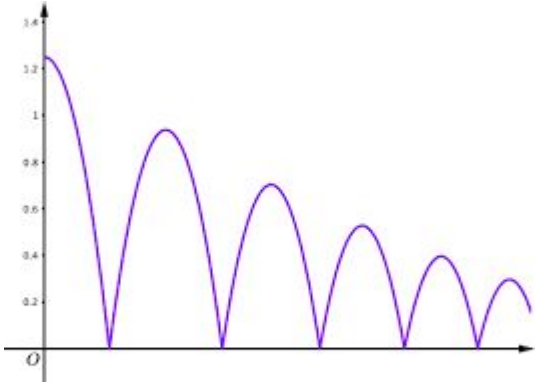


Simulation of Bouncing Ball

A bouncing ball model is an example of a **hybrid dynamic system**.

A hybrid dynamic system is a system that involves both **continuous dynamics** and **discrete transitions** where the system dynamics can change and the state values can jump.





Bouncing Ball Dynamics (P)

Continuous Dynamics

$$\frac{dv}{dt} = -g$$

$$\frac{dx}{dt} = v$$

g: Gravity

v: velocity

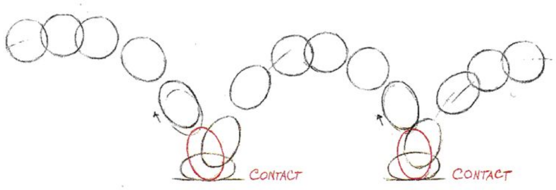
x: position

Discrete Transition

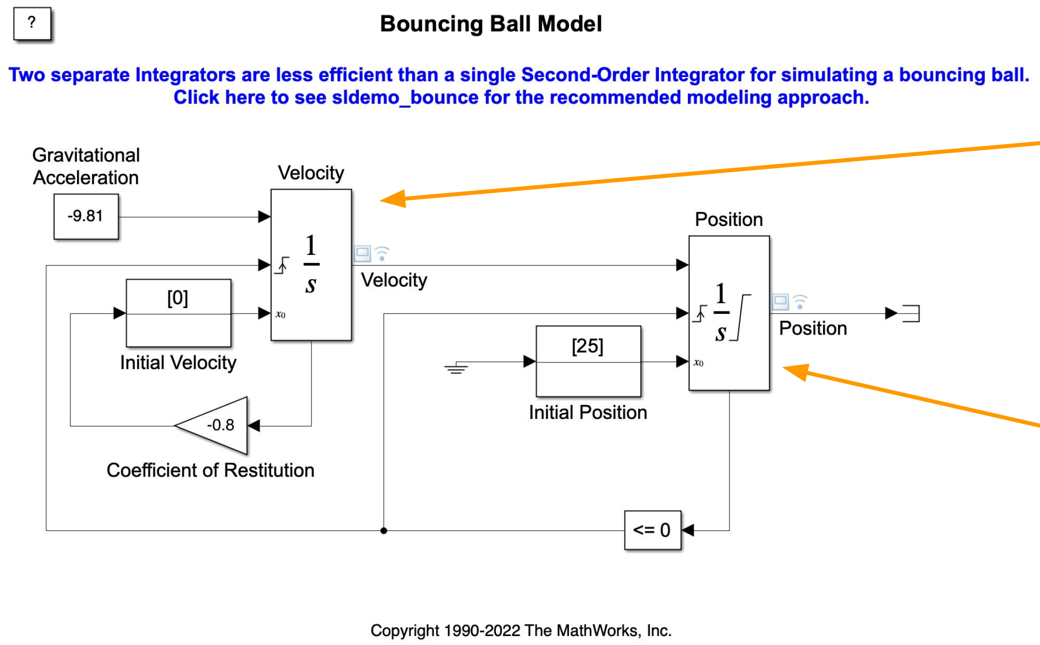
$$v^{+} = -kv^{-}$$

$$x = 0$$

The ball displays a jump at the transition condition x=0



Bouncing Ball Simulink



High-level overview

Left integration block

- Integrate acceleration (gravity) to compute velocity

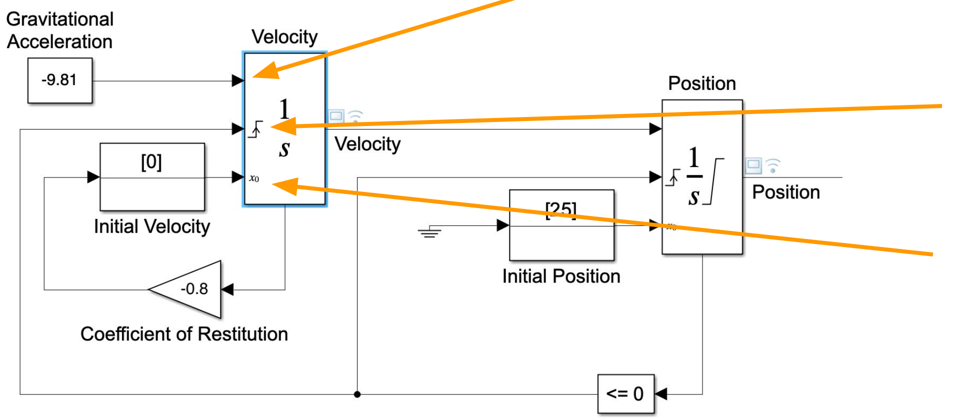
$$\frac{dv}{dt} = -g$$

Right integration block

- Integrate velocity to compute position

$$\frac{dx}{dt} = v$$

Velocity Integrator



Input that we want to integrate $\frac{dv}{dt} = -g$

- Integrate gravitational acceleration over time

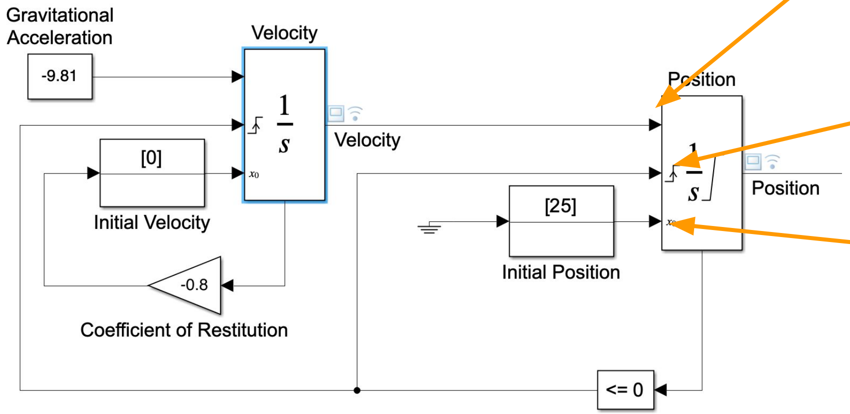
Reset condition $x = 0$

- Reset velocity when Position <= 0

Reset value $v^+ = -kv^-$

- Initial velocity = 0
- If reset condition is met, reset velocity with -0.8v

Position Integrator



Input that we want to integrate $\frac{dx}{dt} = v$

- Integrate velocity over time

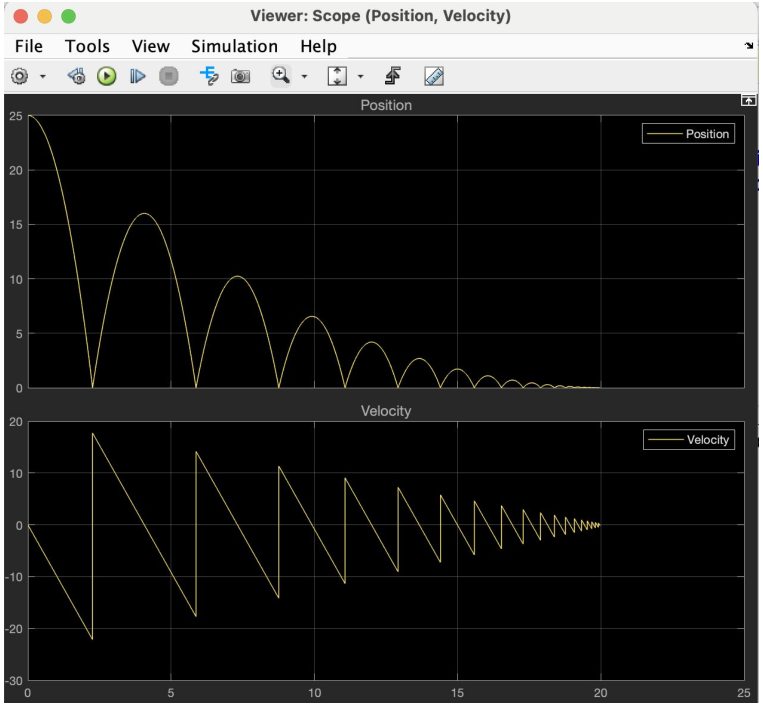
Reset condition

- Reset position when Position <= 0

Reset value

- Initial position = 25
- If reset condition is met, reset to 0

Simulation Result



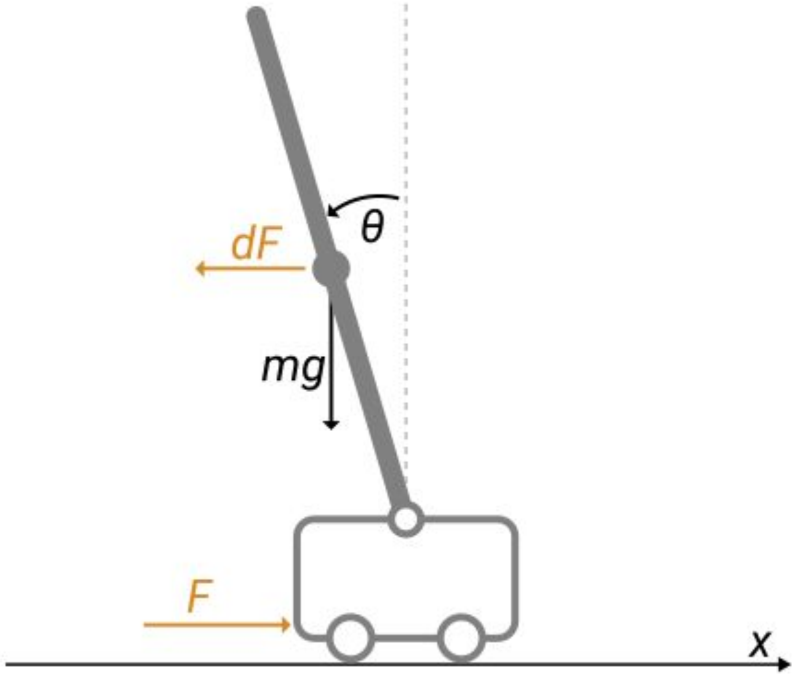
When position = 0, the ball bounces off the ground

Velocity's sign is changed, magnitude got reduced

Eventually, the ball loses the energy during each bounce

```
openExample('simulink_general/sldemo_bounceExample')
```

More Complex Examples



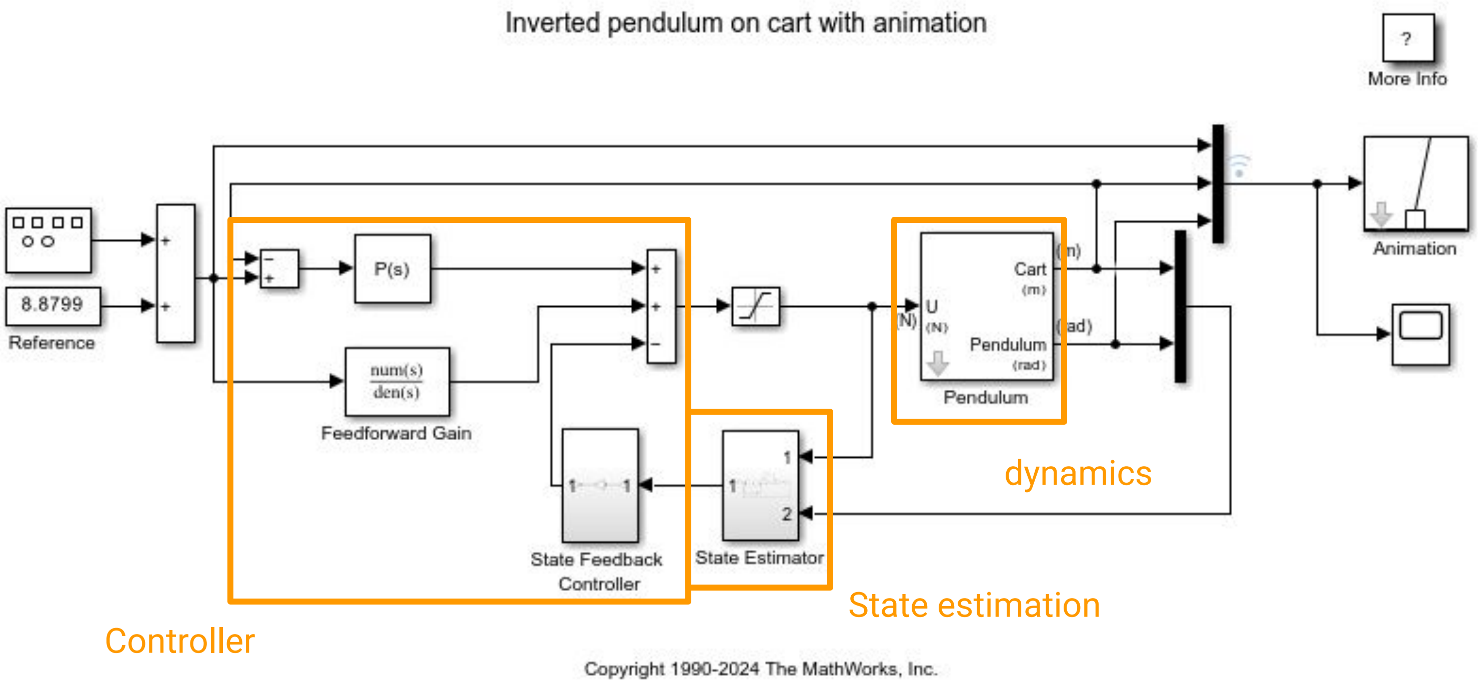
Control of Inverted Pendulum

Classic dynamics and control problem

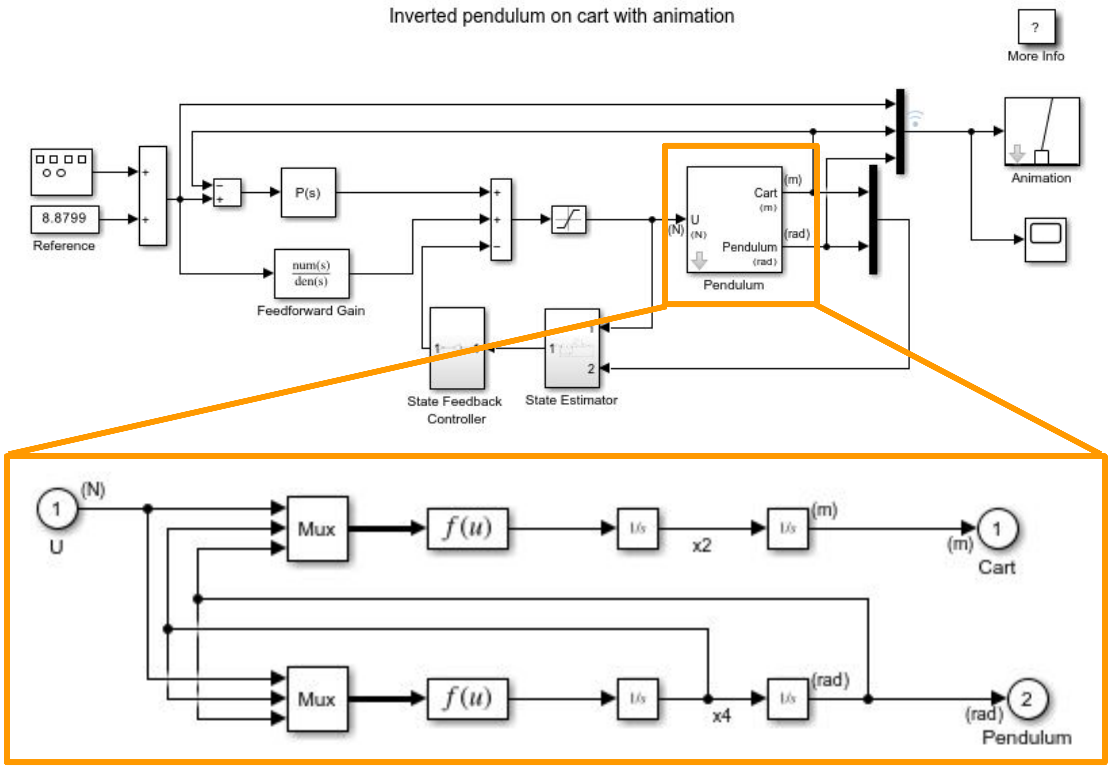
Center of mass above its pivot point - Unstable system

Require control logic to avoid pendulum falling off and maintaining position

Inverted Pendulum



Nested Blocks – Pendulum Dynamics



Animations


SCOPE
MEASUREMENTS
Reference Position
Cart Position
Pendulum Position
Refernce and Cart Position (m), Pendulum Position (r
10
8
6
4
2
0
0
3
6
9
12
15
18
Time (secs)
Ready
Sample based
T=20.0000

References & Resources

Bouncing Ball example: **Simulation of Bouncing Ball**

Pendulum example: **Inverted Pendulum with Animation**

Silverstone Racing Track example:
MW208_Raceline_Optimization